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Statistical and Dimensionality-Reduction Analysis of the Communities and Crime Dataset

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This work was conducted as the final project of the CSE 555: Statistical Data Analysis course in the Master's Degree Program at Gebze Technical University.

ABSTRACT This work investigates the structure of the UCI *Communities and Crime* dataset through a sequence of exploratory, statistical, and dimensionality-reduction analyses. The 122 predictive attributes of the dataset are first reduced to a working set of forty-three normalized sociodemographic variables through a label-blind, data-driven feature selection pipeline: low-coverage LEMAS attributes are dropped, and the remaining predictors are grouped by agglomerative hierarchical clustering on absolute Pearson correlation distance ($d = 1 - |r|$, complete linkage), from each cluster one representative being retained. The continuous target `ViolentCrimesPerPop` is discretized into Low, Medium, and High crime classes by tertiles. The pipeline includes per-feature box-plot inspection and Isolation-Forest-based multivariate outlier cleaning, Pearson and Spearman correlation analysis, z-score normalization with per-feature Fisher Distance, principal component analysis (PCA) with Fisher Distance per principal component, linear discriminant analysis (LDA), and a comparison of K-Means, DBSCAN, t-SNE, and Self-Organizing Map projections. Quantitative separability is reported through silhouette and 5-NN cross-validation accuracy on the 2-D embeddings, and the SOM organisation is summarised by a per-neuron class-purity hit-map. The analysis shows that family-structure and socioeconomic representatives (e.g., `PctKids2Par`, `PctIlleg`, `PctPopUnderPov`) carry the strongest individual class signal, that the leading PCA component inherits most of this discriminative content, and that LDA produces a tighter class-separable 2-D embedding than PCA. A Welch's t-test on `PctKids2Par` between Low and High classes establishes a highly significant mean difference at both $n = 36$ and $n = 64$.

INDEX TERMS Communities and Crime, exploratory data analysis, hierarchical clustering, feature selection, Fisher Distance, PCA, LDA, K-Means, DBSCAN, t-SNE, Self-Organizing Map, hypothesis testing.

I. INTRODUCTION

THE *Communities and Crime* dataset from the UCI Machine Learning Repository [1] is a widely used benchmark for sociodemographic modelling of community-level violent crime in the United States. It aggregates information from the 1990 U.S. Census, the 1990 U.S. Law Enforcement Management and Administrative Statistics (LEMAS) survey, and the 1995 FBI Uniform Crime Reports (UCR) [2], and contains 1 994 communities described by 128 attributes. Each predictive attribute is already normalized to the $[0, 1]$ interval by the dataset authors so that comparisons across communities of different population are meaningful.

The aim of this study is not to build a competitive predictive model, but to characterize the statistical structure of this real, mixed-quality dataset through a sequence of complementary exploratory, dimensionality-reduction, and inferential analyses, and to interpret what each method exposes about the data. A central methodological choice is to replace subjective hand-curation of the working feature set with a fully data-driven, label-blind dimensionality reduction. Concretely, low-coverage attributes are first dropped, then the remaining predictors are grouped by agglomerative hierarchical clustering on the absolute Pearson correlation distance, and one representative per cluster is retained. This procedure produces a working set of 43 sociodemographic features that spans every conceptual block of the original dataset — demographics, ethnic composition, urbanization, income and housing cost, education, employment, poverty, family structure, immigration, housing quality, and residential mobility — while bounding the within-cluster pairwise correlation. The selection mechanism is described in Section II. To cast the problem in a multi-class setting, the continuous response

`ViolentCrimesPerPop` is discretized into three equal-frequency classes *Low*, *Medium*, and *High* by its tertiles.

The remainder of the paper is organized as follows. Section II describes the dataset, the missingness filter, and the dendrogram-based feature selection. Section III examines the feature distributions and performs multivariate outlier removal. Section IV analyzes the correlation structure among features and with the class label, and Section V quantifies the per-feature discriminability through the Fisher Distance. Sections VI–VII study the principal-component structure and its low-dimensional class projections, and Section VIII contrasts these with a supervised linear discriminant embedding. Section IX probes the unsupervised cluster structure with K-Means, DBSCAN, t-SNE, and a Self-Organizing Map, and Section X tests the significance of the leading predictor. Section XI summarizes the main findings.

II. DATASET AND FEATURE SELECTION

The Communities and Crime dataset [1] contains 1 994 communities described by 128 columns: five non-predictive identifiers (`state`, `county`, `community`, `communityname`, `fold`), 122 candidate predictors, and the goal attribute `ViolentCrimesPerPop`. All predictive variables are already normalized to $[0, 1]$ by the dataset authors through an unsupervised equal-interval binning that clips values beyond ± 3 standard deviations; this normalization preserves the within-variable distribution shape but does not preserve relative magnitudes across variables. The continuous response `ViolentCrimesPerPop` is discretized into three equal-frequency classes *Low*, *Medium*, and *High* by its tertiles, yielding 679/662/653 samples before any cleaning. All processing was carried out in Python 3.10 using `NumPy`, `pandas`, `scikit-learn` [6], `scipy`, `seaborn`, and `MiniSom` [15].

A. IDENTIFIER REMOVAL AND MISSINGNESS FILTER

The five identifier columns were dropped first. The dataset then contains a contiguous block of 22 variables derived from the LEMAS police-force survey (e.g., `LemasSwornFT`, `PolicPerPop`, `RacialMatchCommPol`) that is missing for 1 675 of 1 994 communities, i.e., for 84% of the sample. Imputing such a deep gap would inject substantial artefactual variance and contaminate any downstream covariance-based method (PCA, LDA, Fisher Distance); these 22 columns were therefore removed entirely. The remaining 100 predictors carry at most 0.05% missingness each and were imputed by the per-column median.

B. FEATURE SELECTION VIA HIERARCHICAL CLUSTERING ON CORRELATION DISTANCE

The 100 surviving predictors still contain heavy conceptual redundancy: `medIncome`, `medFamInc`, `perCapInc`, `whitePerCap`, and the owner-occupied housing value quartiles `OwnOcc{Low,Med,Hi}Quart` all measure the same underlying construct (community wealth) at pairwise $|r| > 0.85$. A hand-curated working set inevitably reflects the analyst's prior expectations; we therefore replace it with an entirely label-blind procedure based on agglomerative hierarchical clustering [5] of features in their pairwise-correlation geometry. Because the class label is never consulted during selection, the downstream supervised analyses (Fisher Distance, LDA, K-NN separability) remain free of selection-induced data leakage.

Two features that carry the same information must be treated as redundant regardless of the sign of their correlation: `racePctWhite` and `racepctblack` correlate at $r \approx -0.85$ but encode a single ethnic-composition axis. We therefore use the absolute-value Pearson correlation distance

$$d_{ij} = 1 - |r_{ij}|, \qquad d_{ij} \in [0, 1], \tag{1}$$

which maps both perfectly positively and perfectly negatively correlated pairs to $d = 0$ and statistically independent pairs to $d = 1$. The condensed pairwise distance matrix over the 100 predictors is passed to agglomerative clustering with *complete linkage*. Complete linkage merges two clusters only when the *maximum* pairwise distance across them does not exceed the threshold, which guarantees that within every resulting cluster the pairwise correlation is bounded from below by $1 - d_{\mathrm{th}}$. Average and single linkage do not provide this guarantee and produce clusters that mix tightly and loosely correlated members.

A threshold sweep over $d_{\mathrm{th}} \in \{0.20, 0.25, \ldots, 0.60\}$ yields between 57 and 29 clusters, respectively. The operating point $d_{\mathrm{th}} = 0.35$ — corresponding to a within-cluster correlation floor of $|r| \geq 0.65$ — was retained. This choice produces exactly 43 clusters, which is the smallest cluster count at which the four largest obvious redundancy blocks (income/wealth, family-structure, deprivation, and recent-immigration shares) consolidate into single groups, while conceptually distinct dimensions (e.g., immigration recency proportions vs. immigration share of population, or two-parent households vs. never-married mothers) remain separate. Figure 1 shows the resulting dendrogram with the cut and the colour-coded clusters.

C. SELECTION OF A CLUSTER REPRESENTATIVE

From each of the 43 clusters exactly one representative is retained. For singleton clusters the representative is the cluster member itself (24 singletons). For each multi-member cluster, the default rule is to take the *centroid* feature, i.e., the member whose mean pairwise correlation distance to the other members is lowest; this is the feature that best summarizes the cluster on its own. For 7 of the 19 multi-member clusters the centroid coincides with a feature widely used in the criminology and social-science literature and is kept as is. For the other 12 multi-member clusters an interpretability-driven override was applied to favour the canonical literature variable over a near-equivalent centroid (e.g., `medIncome` is preferred to the centroid `RentHighQ` in the income-and-wealth cluster, since median income is the standard community-wealth proxy in the relevant literature). All overrides stay inside the

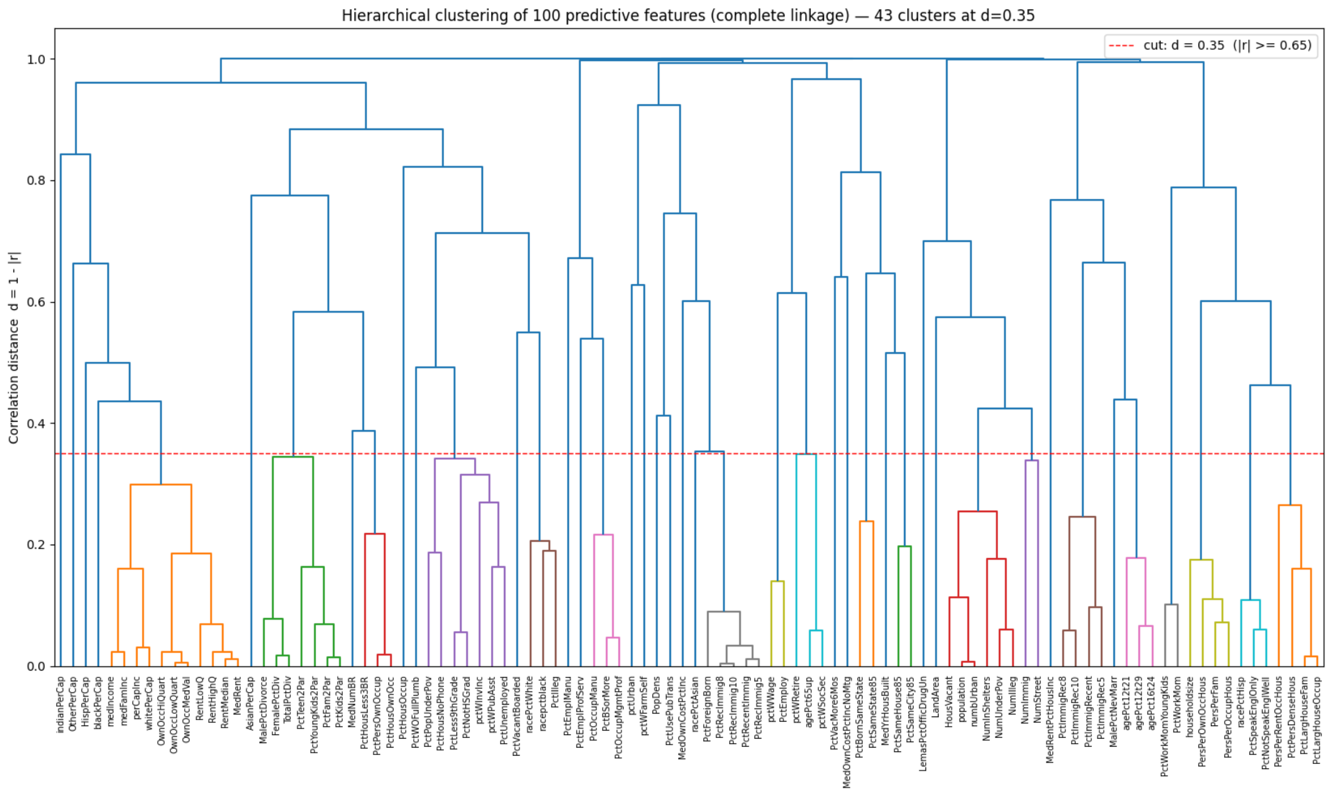

FIGURE 1. Agglomerative hierarchical clustering of the 100 candidate predictors on the absolute Pearson correlation distance $d = 1 - |r|$ with complete linkage. The horizontal red line at $d = 0.35$ is the cut that produces 43 clusters and enforces a within-cluster correlation floor of $|r| \geq 0.65$. Singleton clusters (vertical leaves that reach $d \approx 1$ without joining any sub-tree below the cut) correspond to features that carry conceptually unique information.

cluster they replace, so the within-cluster correlation guarantee is preserved. The multi-member clusters and their final representatives are listed in Table 1.

D. FINAL WORKING SET

The procedure of Sections II-A–II-C produces a deterministic working set of 43 features — 24 from singleton clusters and 19 from multi-member clusters — listed by conceptual block in Table 2. Each retained feature represents a non-redundant axis of variation in the data, in the precise sense that no two retained features sit inside the same hierarchical-clustering cluster at the chosen correlation floor. Inter-cluster correlations are of course not bounded by this procedure: the cluster-12 representative `PctIlleg` and the cluster-6 representative `PctKids2Par`, for instance, still correlate at $|r| \approx 0.87$ because the dendrogram exposes them as two distinct but conceptually proximate "concentrated disadvantage" axes. This residual between-cluster correlation is itself a substantive observation about the dataset and is revisited in Sections IV and VI. The same 43-feature working set is used in every subsequent analysis.

III. FEATURE DISTRIBUTIONS AND OUTLIER REMOVAL

Figure 2 presents per-feature box plots of the 43 selected variables before outlier removal, split into two panels and annotated by conceptual block for readability. Three distributional patterns are apparent. First, several features are strongly right-skewed with a long upper tail of extreme values — most notably `population`, `PopDens`, `LandArea`, `NumImmig`, `PctIlleg`, and `PctVacantBoarded`; these tails reflect a small set of communities that are demographically or socioeconomically extreme (very large or dense cities, very high immigration counts, or concentrated deprivation). Second, the protective family and housing variables (`PctKids2Par`, `PctHousOwnOcc`, `PctEmploy`) are *left*-skewed, with a long lower tail of disadvantaged communities. Third, `pctUrban` is markedly bimodal: its box spans almost the entire $[0, 1]$ range because communities are predominantly either fully urban (≈ 1) or fully rural (≈ 0), a characteristic worth recalling when interpreting the projections of later sections. In contrast, `agePct12t29`, `agePct65up`, and the per-capita income variables are tightly concentrated around their median.

Why a multivariate detector. A per-feature filter such as univariate IQR rejection discards any sample that is extreme on *any* single dimension. This is doubly inappropriate here. Conceptually, several class-discriminative features (`PctPopUnderPov`, `PctIlleg`, `PctVacantBoarded`) intrinsically carry heavy upper tails populated almost exclusively by High-crime communities, so marginal trimming would deplete precisely the class the analysis is meant to characterize. Statistically, the probability

TABLE 1. **The 19 multi-member clusters at $d_{\text{th}} = 0.35$, their members, and the representative retained for downstream analysis. "$\star$" marks the algorithmic centroid; the chosen representative is shown in bold. Singleton clusters (24 features) are listed in Table 2.**

Cluster	Size	Theme	Members (centroid $\star$, representative in **bold**)
1	11	Income / wealth	**medIncome**, medFamInc, perCapInc, whitePerCap, OwnOcc{Low,Med,Hi}Quart, Rent{Low,Med}Q, RentHighQ*, MedRent
6	7	Family structure	MalePctDivorce, FemalePctDiv, TotalPctDiv, PctFam2Par*, **PctKids2Par**, PctYoungKids2Par, PctTeen2Par
7	3	Home ownership	PctPersOwnOccup*, PctHousLess3BR, **PctHousOwnOcc**
10	7	Deprivation	pctWInvInc, pctWPubAsst, **PctPopUnderPov**, PctLess9thGrade, PctNotHSGrad*, PctUnemployed, PctHousNoPhone
12	3	Race / illegitimacy	racepctblack, racePctWhite, **PctIlleg***
15	3	Education / occupation	**PctBSorMore**, PctOccupManu, PctOccupMgmtProf*
22	5	Immigration share	PctRecentImmig, PctRecImmig5, PctRecImmig8*, PctRecImmig10, **PctForeignBorn**
25	2	Employment	pctWWage*, **PctEmploy**
26	3	Elderly / retirement	**agePct65up***, pctWSocSec, pctWRetire
29	2	State-level residence	**PctBornSameState***, PctSameState85
30	2	Local residence	**PctSameHouse85***, PctSameCity85
32	6	Population-scaled counts	**population***, numbUrban, NumUnderPov, NumIlleg, HousVacant, NumInShelters
33	2	Population-scaled counts	**NumImmig***, NumStreet
36	4	Immigration recency	PctImmigRecent, PctImmigRec5, **PctImmigRec8***, PctImmigRec10
37	3	Youth age structure	agePct12t21, **agePct12t29**, agePct16t24*
40	2	Working motherhood	PctWorkMomYoungKids*, **PctWorkMom**
41	4	Household size	**householdsize**, PersPerFam, PersPerOccupHous*, PersPerOwnOccHous
42	3	Hispanic / language	**racePctHisp**, PctSpeakEnglOnly, PctNotSpeakEnglWell*
43	4	Crowded housing	PctLargHouseFam, PctLargHouseOccup*, PersPerRentOccHous, **PctPersDenseHous**

TABLE 2. **The 43 retained features grouped by conceptual block. All variables are already normalized to [0, 1] by the dataset authors.**

Block	Features
Demography	population, householdsize, agePct12t29, agePct65up, racePctAsian, racePctHisp
Urbanization	pctUrban, PopDens, PctUsePubTrans, LandArea
Income / wealth	medIncome, blackPerCap, HispPerCap, AsianPerCap, indianPerCap, OtherPerCap
Housing cost	MedOwnCostPctInc, MedOwnCostPctIncNoMtg, MedRentPctHousInc
Education	PctBSorMore
Employment	PctEmploy, PctEmplProfServ, PctEmplManu, pctWFarmSelf
Poverty	PctPopUnderPov
Family	PctKids2Par, PctIlleg, MalePctNevMarr, PctWorkMom
Immigration	NumImmig, PctForeignBorn, PctImmigRec8
Housing quality	PctHousOwnOcc, MedNumBR, PctVacantBoarded, PctHousOccup, PctVacMore6Mos, MedYrHousBuilt, PctWOFullPlumb, PctPersDenseHous
Mobility	PctBornSameState, PctSameHouse85
Policing	LemasPctOfficDrugUn

that a row is flagged grows rapidly with the feature count, because a row is rejected as soon as it exceeds the threshold on *at least one* of the 43 marginals — a direct manifestation of the curse of dimensionality. Table 3 quantifies both effects: a per-feature IQR rule with multiplier $k = 3$ removes 730 rows (36.6% of the data) and collapses the High-crime class by more than half ($653 \to 295$, -54.8%), which would render

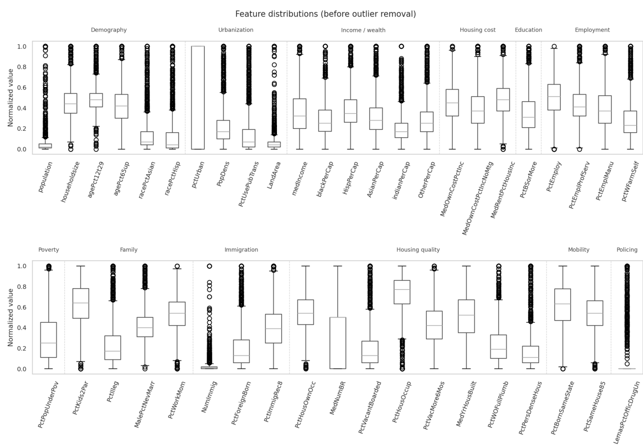

FIGURE 2. **Box plots of the 43 selected features prior to outlier removal, grouped by conceptual block. Several socioeconomic variables show clear right-skew with long upper tails, the family/housing protective variables are left-skewed, and** `pctUrban` **is bimodal.**

the supervised analyses of Sections V–VIII meaningless.

The final pipeline therefore uses the multivariate *Isolation Forest* [7] of `scikit-learn` [6] with 200 trees and a contamination rate of 5%. The detector isolates each sample by recursive random partitioning of the feature space; a sample that requires unusually few random splits to be isolated occupies a sparse region of the 43-dimensional space and is flagged as a *joint* anomaly. With contamination $= 0.05$ the filter removes 100 rows (5.0%), leaving 1 894 communities with class distribution *Low*: 660, *Medium*: 649, *High*: 585

TABLE 3. Outlier-cleaning comparison on the 43-feature working set. The per-feature IQR rule is shown only as a rejected baseline; the multivariate Isolation Forest is the method actually used.

	Before	IQR ($k = 3$)	Isolation Forest
Rows kept	1 994	1 264	1 894
Rows removed	—	730 (36.6%)	100 (5.0%)
Low	679	524 (−22.8%)	660 (−2.8%)
Medium	662	445 (−32.8%)	649 (−2.0%)
High	653	295 (−54.8%)	585 (−10.4%)

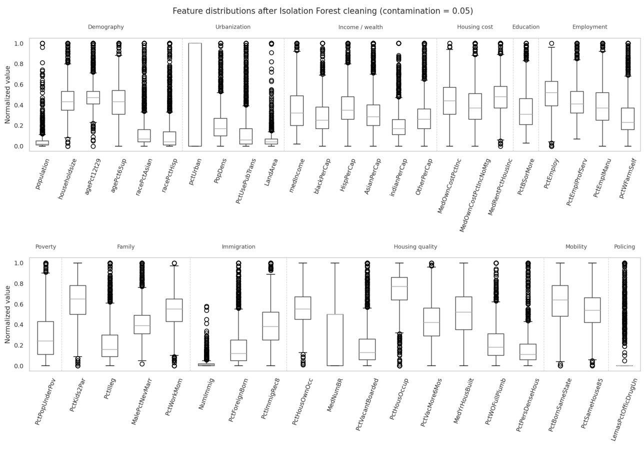

FIGURE 3. Box plots after Isolation Forest cleaning (contamination = 0.05). The bulk of each distribution is preserved, yet marginal outlier points (circles beyond the whiskers) remain visible — a direct visual confirmation that the detector removes joint anomalies rather than per-feature extremes.

(Table 3). Notably, the High-class shrinkage is *milder* on the 43-feature set (−10.4%) than on a smaller 18-feature set (−11.6% in preliminary runs): the additional dimensions give the detector more evidence with which to separate legitimately extreme High-crime communities from communities that are genuinely anomalous across many axes simultaneously.

A subtle but important confirmation is visible in Figure 3: many marginal outlier points (circles beyond the whiskers) *persist* after cleaning. A per-feature IQR rule, by construction, would have produced boxplots with no points beyond the whiskers; their persistence here demonstrates that Isolation Forest targets samples that are anomalous in the *joint* 43-dimensional distribution, not those that are merely extreme on an individual axis. The qualitative shape of every feature distribution — skew, median, and interquartile range — is preserved, and all subsequent analyses use this cleaned dataset of 1 894 communities.

IV. CORRELATION STRUCTURE

The full Pearson correlation matrix among the 43 features and the ordinally encoded class label is shown in Figure 4. With 44 rows the per-cell numeric annotations would be illegible, so they are omitted and the conceptual-block boundaries of Table 2 are drawn as black separators; the colour field alone then makes the macro-structure legible. The same two opposed super-blocks that the dendrogram of Section II exposed re-appear here in correlation space. An *advantage* super-block — median income and the per-capita income variables, education (`PctBSorMore`), employment (`PctEmploy`), family stability (`PctKids2Par`), and home ownership (`PctHousOwnOcc`, `MedNumBR`) — is internally positively correlated. A *disadvantage* super-block — poverty (`PctPopUnderPov`), illegitimacy (`PctIlleg`), overcrowding (`PctPersDenseHous`), and housing decay (`PctVacantBoarded`) — is likewise internally positively correlated. The two super-blocks are strongly anti-correlated with each other, which is the well-known socioeconomic "advantage–disadvantage" gradient that the rest of this paper repeatedly recovers.

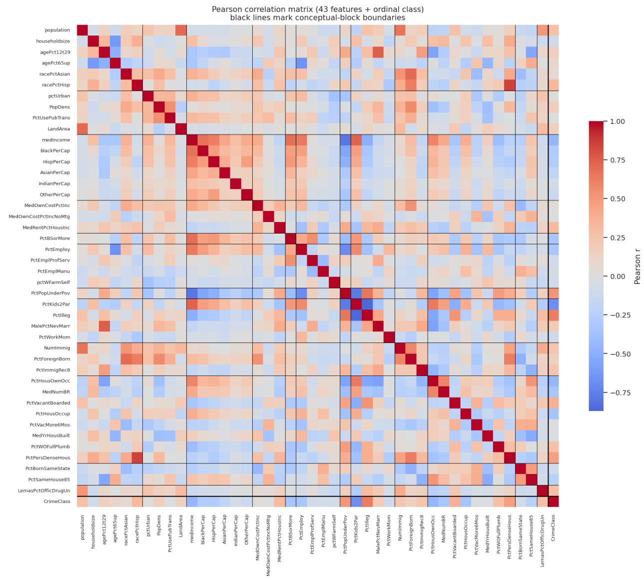

FIGURE 4. Pearson correlation matrix of the 43 features and the ordinally coded crime class. Numeric annotations are omitted for legibility; black lines mark the conceptual-block boundaries of Table 2. The opposed warm/cool blocks are the advantage–disadvantage socioeconomic gradient.

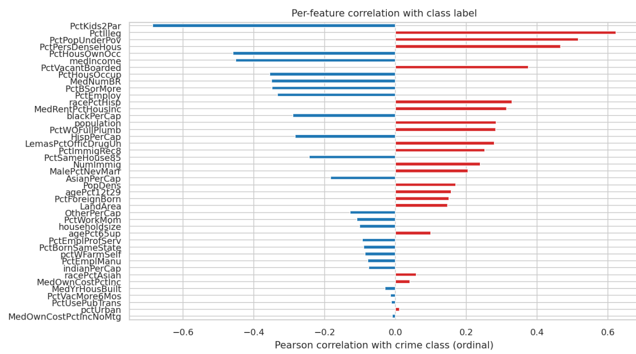

FIGURE 5. Per-feature Pearson correlation with the (ordinal) crime class, all 43 features ranked. Red bars indicate positive correlation with higher crime levels, blue bars negative (protective) correlation.

Figure 5 ranks all 43 features by their correlation with the (ordinal) crime class. The strongest individual correlates are `PctKids2Par` (-0.69), `PctIlleg` ($+0.62$), `PctPopUnderPov` ($+0.52$), `PctPersDenseHous` ($+0.47$), `PctHousOwnOcc` (-0.46), `medIncome` (-0.45), and `PctVacantBoarded` ($+0.38$). The signs are consistent with the social-disorganization account of community-level crime [3], [4]: family stability, income, education, and stable owner-occupied housing are protective, whereas poverty, family disruption, and housing decay co-occur with higher violent crime. Two observations are specific to the data-driven 43-feature set. First, `PctPersDenseHous` (the share of people living in housing with more than one occupant per room) emerges as the fourth-strongest individual correlate; this overcrowding signal is conceptually distinct from poverty and would have been easy to overlook in a smaller hand-picked set, yet the clustering pipeline retains it as the representative of its own block. Second, the housing-quality block is collectively informative rather than incidental: `PctHousOwnOcc`, `PctVacantBoarded`, `PctHousOccup`, and `MedNumBR` all rank in the top ten, indicating that the physical state of the housing stock tracks community crime almost as closely as the canonical income and family variables. At the opposite end, demographic and geographic variables — `pctUrban`, `LandArea`, `agePct65up`, `householdsize`, `MedYrHousBuilt`, and the ethnicity-specific per-capita income variables — carry essentially no individual class signal.

Because the class label is an *ordinal* tertile rather than an interval-scale variable, the Spearman rank correlation was also computed for the feature–class relationship. The Spearman magnitudes are typically slightly larger (e.g., `PctKids2Par` -0.70, `PctIlleg` $+0.69$, `PctPopUnderPov` $+0.57$). The most pronounced shift is `PctPersDenseHous`, which rises from $+0.47$ (Pearson) to $+0.61$ (Spearman, third overall); the gap indicates that overcrowding relates to the crime tertile monotonically but non-linearly, consistent with a threshold effect in which crime rises sharply only once density passes a critical level. Despite such individual shifts, the absolute-correlation *ranking* of features under the two metrics is almost identical (Spearman rank agreement 0.97), confirming that the qualitative conclusions of this section are not artefacts of treating the ordinal class label as interval-scaled.

V. PER-FEATURE FISHER DISCRIMINABILITY

After z-scoring, the per-feature Fisher Distance was computed as a multi-class generalisation of the classical two-class Fisher score [8]:

$$J(f) = \frac{\sum_{c=1}^{C} n_c\,(\mu_c^{(f)} - \mu^{(f)})^2}{\sum_{c=1}^{C}\sum_{i\in c}(x_i^{(f)} - \mu_c^{(f)})^2}, \qquad (2)$$

where $\mu_c^{(f)}$ is the class-c mean of feature f and $\mu^{(f)}$ is its global mean. Larger $J(f)$ indicates that feature f separates the class means well relative to within-class dispersion.

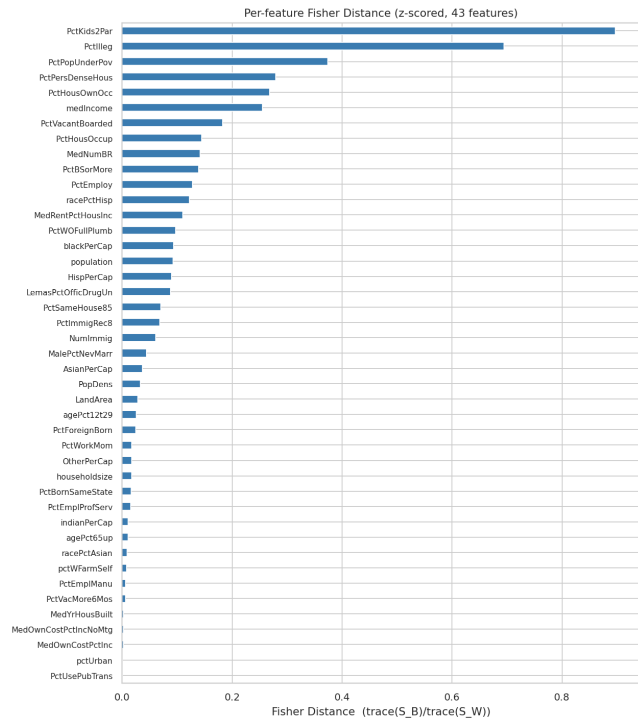

FIGURE 6. Per-feature Fisher Distance after z-score normalization, all 43 features ranked. `PctKids2Par` **and** `PctIlleg` **dominate; roughly half of the features have Fisher Distance close to zero.**

The resulting ranking (Figure 6) is almost identical to the correlation ranking of Section IV, which is expected: for a single z-scored feature, both the Fisher Distance and the absolute class correlation measure the contrast between class means relative to within-class scatter. The two top-ranked features are `PctKids2Par` ($J = 0.90$) and `PctIlleg` ($J = 0.69$), followed by `PctPopUnderPov` (0.37), `PctPersDenseHous` (0.28), `PctHousOwnOcc` (0.27), and `medIncome` (0.26). Two structural observations stand out. First, the discriminative content is sharply concentrated: `PctKids2Par` and `PctIlleg` dwarf every other feature, and there is a large drop to the third-ranked `PctPopUnderPov`, whose score is less than half that of `PctIlleg`. Second, roughly half of the 43 features have a Fisher Distance below 0.03 — `pctUrban` and `PctUsePubTrans` are effectively zero ($J \approx 2 \times 10^{-4}$), as are the housing-cost ratios `MedOwnCostPctInc` and `MedYrHousBuilt`. In other words, although the working set spans 43 conceptually distinct axes, the single-feature class signal lives in only about six to ten of them. This concentration foreshadows the principal-component analysis of Section VI: when most individual axes are class-uninformative and the informative ones are mutually correlated, a very small number of linear combinations should capture essentially all of the separability.

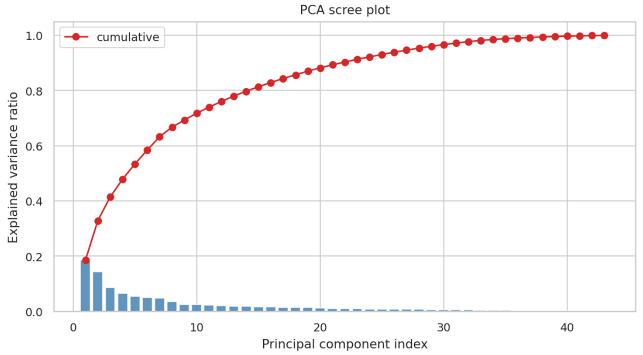

FIGURE 7. PCA scree plot for the 43-feature working set. PC1 captures 18.6% of the variance and PC2 a further 14.3%; the top five components capture 53.4%, and sixteen are needed for 80%.

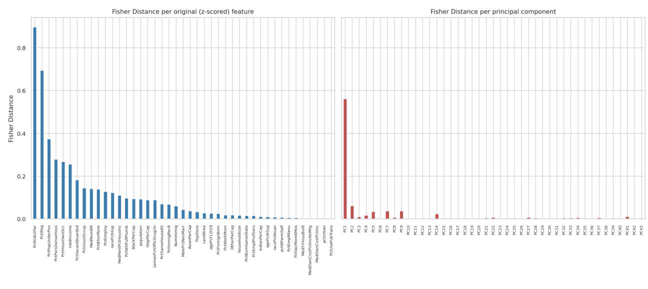

FIGURE 8. Fisher Distance per z-scored original feature (left) and per principal component (right), on a shared vertical scale. The per-feature signal is spread over several variables, whereas the per-PC signal is concentrated almost entirely in PC1.

VI. PRINCIPAL COMPONENT ANALYSIS

PCA [9] was applied to the z-scored data, producing 43 principal components. The leading component PC1 explains 18.6% of the variance and PC2 a further 14.3%; the first five components together account for only 53.4% of the total variance, and sixteen components are needed to reach 80% (Figure 7). The scree curve is therefore far flatter than it would be for a small hand-picked feature set: because the 43 features deliberately span many weakly-correlated conceptual blocks, the variance is distributed across many directions rather than collapsing onto a single dominant axis.

Treating each PC projection as a feature, the Fisher Distance from Equation (2) was recomputed (Figure 8). The Fisher Distance of PC1 is 0.56, comfortably the largest among the principal components; every other PC scores below 0.07. Crucially, however, PC1's 0.56 is itself *lower* than the two best original features (`PctKids2Par` at 0.90 and `PctIlleg` at 0.69). The unsupervised leading component is thus a worse single-axis class discriminator than simply retaining the best raw feature, because PC1 is built to maximize total variance and necessarily blends the discriminative "advantage–disadvantage" direction with correlated but less discriminative variation. This is the central motivation for the supervised projection (LDA) of Section VIII.

The relationship between the eigenvalue of a principal component (its variance) and its Fisher Distance is positive

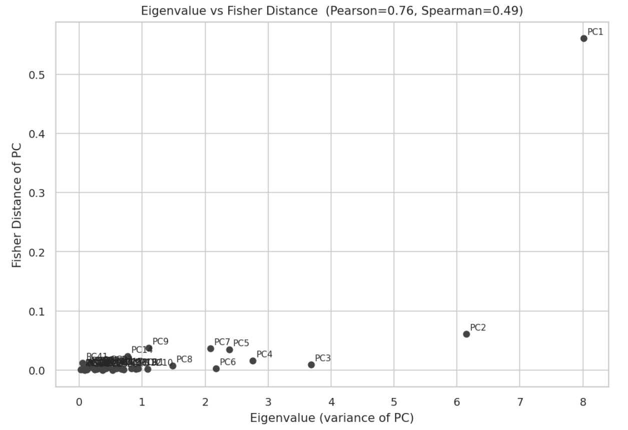

FIGURE 9. Eigenvalue versus Fisher Distance for the 43 principal components. Pearson correlation $r = 0.76$, Spearman $\rho = 0.49$. PC2 has the second-largest eigenvalue yet a near-zero Fisher Distance.

but weak and entirely concentrated in the leading component: the Pearson correlation is $r = 0.76$ and the Spearman rank correlation is only $\rho = 0.49$ (Figure 9). The high Pearson coefficient is an artefact driven solely by PC1, which sits in the far corner of the scatter with both the largest eigenvalue and by far the largest Fisher Distance; with PC1 removed, the eigenvalue–Fisher association essentially vanishes. The clearest illustration is **PC2**: it carries the second-largest variance (14.3%) yet a near-zero Fisher Distance (0.06), and PC3 (third-largest variance) scores only 0.01. These high-variance, class-irrelevant components are a direct demonstration of the textbook caveat that PCA maximizes variance, *not* class separability: PC2 captures a large axis of community variation — orthogonal to the crime gradient — along which the three classes are essentially indistinguishable. Only PC1 happens to align with the discriminative direction, and even it does so imperfectly. This is a more nuanced picture than a smaller feature set would have suggested, where PC1 alone tends to dominate both variance and separability; on the full 43-dimensional set, variance and discriminability are clearly dissociated beyond the leading component.

VII. LOW-DIMENSIONAL CLASS PROJECTIONS

Figure 10 shows the projection of the cleaned data onto the two most important and the two least important principal components, with samples coloured by their true class label.

The (PC1, PC2) plot makes the dissociation of Section VI directly visible. Along the horizontal PC1 axis the three classes form a clean, almost monotone gradient: *Low*-crime communities concentrate on the negative side, *High*-crime communities on the positive side, and *Medium* communities occupy an overlapping band in between. Along the vertical PC2 axis, by contrast, the three classes are thoroughly intermixed: despite carrying the second-largest variance (14.3%), PC2 provides no class separation whatsoever. The entire vertical spread of the plot is therefore class-irrelevant variation — the visual counterpart of PC2's near-zero Fisher Distance

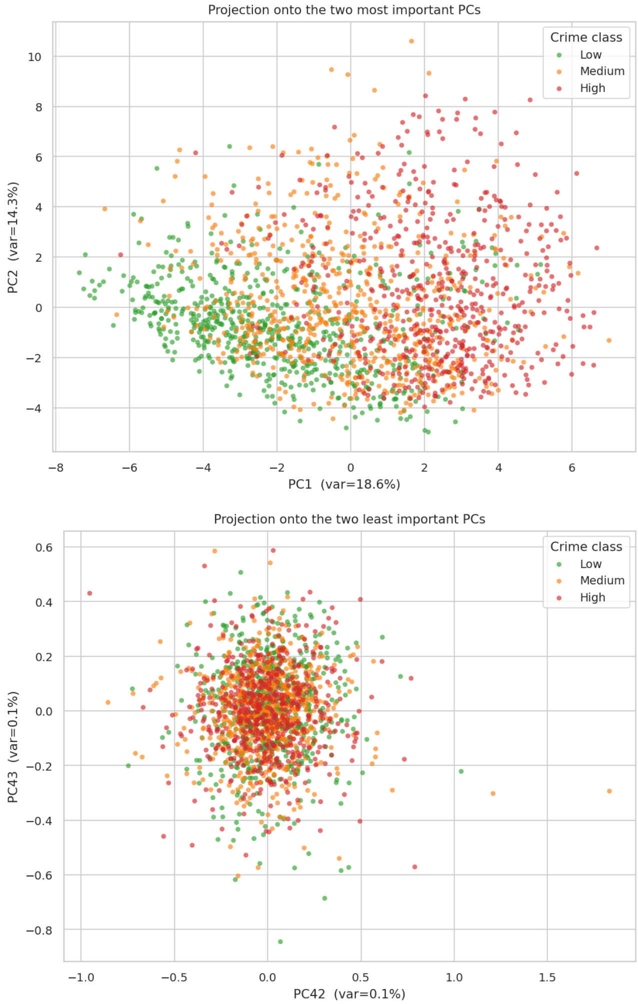

FIGURE 10. **Top: projection onto (PC1, PC2). A clear monotone ordering Low → Medium → High emerges along the horizontal PC1 axis, while the vertical PC2 axis adds no separation. Bottom: projection onto the two least important PCs (PC42, PC43), which show no class structure.**

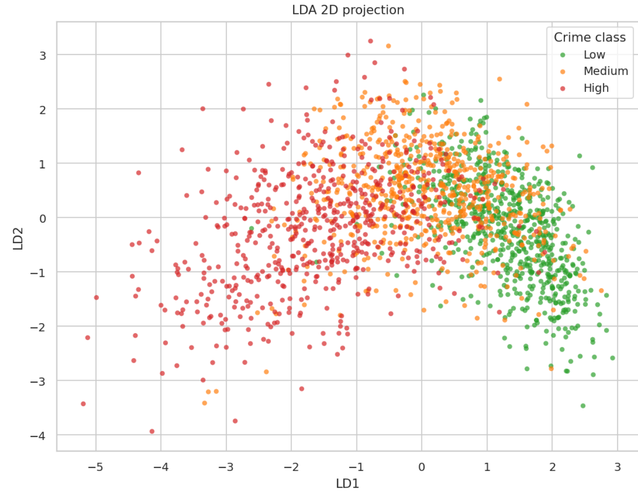

FIGURE 11. **LDA 2-D projection. The three classes form a clearer left–right ordering than in the PCA projection.**

TABLE 4. **Quantitative separability of the 2-D PCA and LDA embeddings. Higher is better in both cases.**

Metric	PCA (PC1,PC2)	LDA (LD1,LD2)
Silhouette (true labels)	0.049	0.127
5-NN accuracy (5-fold CV)	0.569	0.659

— and confirms that what makes a community high-crime is its position along the single PC1 "advantage–disadvantage" axis, not its position on the largest competing axis of variation. The projection onto the two least important components (PC42, PC43), each capturing only 0.1% of the variance, shows the opposite extreme: the three classes are interleaved in a small isotropic cloud with no discernible structure, as expected for directions that capture noise-like residual variance.

VIII. LINEAR DISCRIMINANT ANALYSIS AND CLASS SEPARABILITY

A Linear Discriminant Analysis [8] was fit with $n_{\text{components}} = 2$. By construction, LDA maximizes between-class scatter relative to within-class scatter and therefore tends to produce a tighter class-separable 2-D embedding than PCA when class information is available.

To make the comparison quantitative, two metrics were computed on the 2-D embeddings (Table 4). The silhouette coefficient [10] (using the true class labels as the partition) rises from 0.049 for PCA to 0.127 for LDA, an increase of roughly 160% ($2.6\times$). The mean 5-fold cross-validated accuracy of a 5-NN classifier trained on the 2-D embedding improves from 56.9% on PCA to 65.9% on LDA, a gain of $+9.0$ percentage points. Both metrics confirm what Figures 10 and 11 show: LDA reveals substantially tighter class separability than PCA. Importantly, this gap is *wider* on the 43-feature set than a low-dimensional analysis would suggest, and the reason follows directly from Section VI. The PCA embedding must spend one of its two axes (PC2) on the largest available direction of variance, which Section VI showed to be class-irrelevant; only PC1 carries class signal, so PCA effectively has a single useful discriminative axis. LDA, being supervised, orients *both* of its axes to maximize between-class relative to within-class scatter, and therefore exploits the rich 43-dimensional feature space far more efficiently. In short, PCA wastes half of its 2-D budget on class-irrelevant variance whereas LDA does not, and the metric gap quantifies exactly that difference.

IX. UNSUPERVISED CLUSTER STRUCTURE

The four algorithms were applied with two different inputs. K-Means ($k = 3$) [11] and DBSCAN ($\varepsilon = 0.5$, $\text{minPts} = 10$) [12] operate on the 2-D PCA projection (PC1, PC2). t-SNE (perplexity 30, PCA initialization) [13] and a 12×12 Self-Organizing Map [14], [15] are fit on the full 43-dimensional z-scored data.

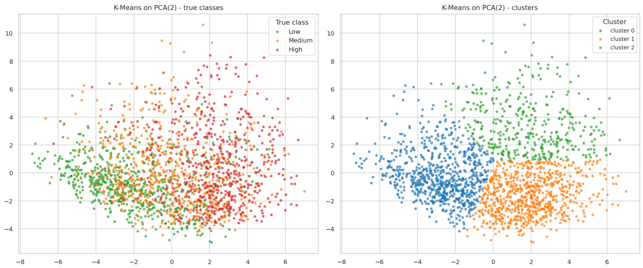

FIGURE 12. K-Means ($k = 3$) on the (PC1, PC2) projection. Left: true classes; right: assigned clusters.

FIGURE 13. DBSCAN ($\varepsilon = 0.5$, minPts = 10) on (PC1, PC2). The bulk of the data collapses into one dominant dense cluster (plus a few tiny fringe clusters), with 233 samples (12.3%) flagged as noise.

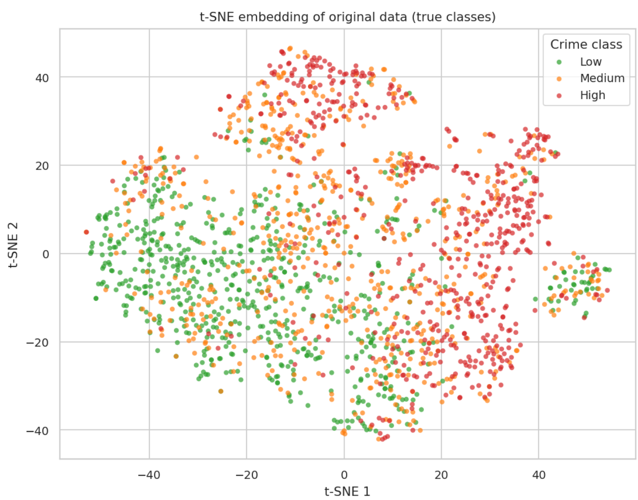

FIGURE 14. t-SNE embedding of the full 43-dimensional z-scored data, coloured by true class. A continuous gradient is visible across the embedding.

TABLE 5. Adjusted Rand Index (ARI) [16] and Normalised Mutual Information (NMI) between cluster assignments and the true class labels.

Method (input)	ARI	NMI
K-Means, $k = 3$ (PCA-2D)	0.142	0.147
DBSCAN ($\varepsilon = 0.5$) (PCA-2D)	0.018	0.039

a: K-Means

on the (PC1, PC2) plane recovers three clusters whose boundaries are roughly vertical cuts along PC1, echoing the gradient of Section VII. Because PC1 is the only one of the two input

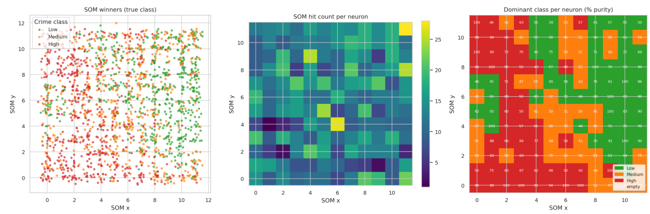

FIGURE 15. 12 × 12 MiniSom trained on the original z-scored data. Left: scatter of winners coloured by true class (jittered for visibility). Centre: per-neuron hit-count heat-map. Right: dominant class per neuron with per-cell purity (text labels, in %); the mean purity of non-empty cells is 67.5%.

axes that carries class signal, partitioning along it gives some agreement with the true labels (ARI $= 0.14$, NMI $= 0.15$), but the agreement is weaker than a small-feature analysis would yield. The reason is the same dissociation identified in Sections VI–VIII: K-Means treats PC1 and PC2 isotropically, yet PC2 (14.3% of variance) is class-irrelevant, so a substantial part of the partitioning effort is spent splitting communities along a direction unrelated to crime. Unsupervised clustering in the PCA plane therefore inherits the weakness of PCA itself.

b: DBSCAN

essentially fails: at $\varepsilon = 0.5$, $\mathrm{minPts} = 10$ it produces one dominant dense cluster, a handful of tiny fringe clusters, and 233 noise points (12.3%), for an ARI of only 0.018. Because the data form a single continuous "socioeconomic gradient" rather than well-separated density modes, density-based clustering cannot recover the three classes; DBSCAN exposes that the class structure of this dataset is gradient-like, not modal.

c: t-SNE

on the full 43-dimensional data produces a single, smoothly varying point cloud along which the three classes form a continuous gradient (*Low* on one side, *High* on the other) rather than separated islands. This reinforces the picture that the classes are largely defined by position along a one-dimensional socioeconomic continuum, not by compact modal clusters — the same conclusion reached independently by PCA, DBSCAN, and the SOM.

d: SOM

The 12 × 12 MiniSom grid (Figure 15) provides the most class-organised non-linear view. The hit-count heat-map (centre panel) shows that the SOM allocates capacity smoothly across the grid with a few slightly heavier hubs. The dominant-class panel (right) reveals contiguous regions of Low, Medium, and High classes arranged along a gradient on the lattice, with several cells reaching $\geq 90\%$ purity. The mean per-neuron purity over non-empty cells is 67.5%, well above the $\approx 35\%$ expected from chance under the (near-uniform) class distribution. The SOM thus confirms, on the

TABLE 6. Welch's t-test results on `PctKids2Par` between the Low and High crime classes.

n per class	$\bar{x}_{\text{Low}}$ ($\pm$ s.d.)	$\bar{x}_{\text{High}}$ ($\pm$ s.d.)	t (p-value)
36	0.759 ± 0.130	0.463 ± 0.147	9.01 (2.8×10^{-13})
64	0.792 ± 0.112	0.444 ± 0.188	12.73 (7.3×10^{-23})
all	–	–	39.28 ($\approx 10^{-207}$)

original 43-dimensional feature space, the same gradient-like organisation seen in the t-SNE map.

X. SIGNIFICANCE OF THE LEADING PREDICTOR

The feature `PctKids2Par` (percentage of kids living in two-parent households) was selected for the hypothesis test because it has the largest Fisher Distance and the largest absolute correlation with the class label (Sections IV–V). The hypothesis to be tested is the equality of its mean between the *Low* and *High* crime classes:

$$H_0 : \mu_{\text{Low}} = \mu_{\text{High}}, \qquad H_1 : \mu_{\text{Low}} \neq \mu_{\text{High}}.$$

Two independent random samples of sizes $n = 36$ and $n = 64$ were drawn (without replacement) from each class, and a two-sided Welch's t-test [17] (which does not assume equal variances) was applied. The results are summarized in Table 6.

In all three settings (samples of size 36, 64, and the full classes) the p-value is many orders of magnitude smaller than any conventional significance level $\alpha \in \{0.05, 0.01, 0.001\}$. The null hypothesis of equal means is therefore rejected with overwhelming evidence in favour of H_1: `PctKids2Par` is, on average, substantially higher in Low-crime communities than in High-crime communities (approximately 0.79 vs. 0.45 on the normalized scale, a ≈ 0.33 shift on a $[0, 1]$ scale). As expected, increasing the sample size from 36 to 64 leaves the sample means almost unchanged but increases $|t|$ from 9.01 to 12.73, illustrating the higher statistical power of a larger sample for the same effect size. The full-class $t = 39.28$ extrapolates this trend to the limit of the available data.

XI. CONCLUSION

This study applied a complete exploratory and statistical analysis pipeline to the UCI Communities and Crime dataset. A label-blind, hierarchical-clustering feature selection reduced the 122 predictors to a non-redundant working set of 43 features, and the continuous violent-crime rate was discretized into three classes. The following coherent picture emerges. First, family-structure and socioeconomic representatives (`PctKids2Par`, `PctIlleg`, `PctPopUnderPov`, `PctPersDenseHous`, `PctHousOwnOcc`, `medIncome`) carry the strongest individual class signal, as confirmed independently by both Pearson and Spearman correlation and by per-feature Fisher Distance (Spearman–Pearson rank agreement 0.97); the data-driven set also surfaces overcrowding (`PctPersDenseHous`) as a top correlate that a hand-picked set might have missed, while demographic and geographic variables such as `pctUrban`, `agePct65up`, and `LandArea` are essentially class-uninformative. Second, the discriminative content is concentrated in very few directions but no longer in a single dominant one: PC1 explains only 18.6% of the variance and obtains a Fisher Distance of 0.56 — itself lower than the two best raw features — while the second-largest variance component PC2 (14.3%) is class-irrelevant (Fisher 0.06). Eigenvalue and Fisher Distance are thus only weakly related ($r = 0.76$ driven entirely by PC1, $\rho = 0.49$): high variance does not imply high discriminability, a point made starkly by PC2. Third, the LDA 2-D embedding clearly outperforms the PCA 2-D embedding in revealing class separability, both visually and quantitatively (silhouette 0.127 vs. 0.049; 5-NN CV accuracy 65.9% vs. 56.9%), precisely because PCA must spend one of its two axes on class-irrelevant variance whereas supervised LDA does not. Fourth, unsupervised clustering reveals that the class structure is gradient-like rather than modal: K-Means on the PCA plane recovers only modest agreement (ARI 0.14), DBSCAN collapses to one dominant cluster with 233 noise points (ARI 0.02), and both t-SNE and SOM produce a continuous manifold along which the three classes shade smoothly into each other; nevertheless the SOM achieves a mean per-neuron class purity of 67.5%, well above the chance level. Finally, a Welch's t-test on `PctKids2Par` rejects the null hypothesis of equal class means with extreme significance at both $n = 36$ ($t = 9.01$) and $n = 64$ ($t = 12.73$), formally confirming the descriptive evidence of the earlier sections.

Overall, the dataset is best understood not as three well-separated clusters but as a one-dimensional socioeconomic continuum along which violent crime rises monotonically. This explains why a supervised linear method (LDA) outperforms the unsupervised PCA projection, why density-based clustering fails, and why the strongest individual predictors are family-structure and deprivation indicators rather than purely demographic ones. The data-driven 43-feature selection both removes the subjectivity of manual curation and exposes structure — such as the dissociation between PCA variance and class separability, and the role of overcrowding — that a smaller hand-picked working set would have obscured.

APPENDIX. A

The Jupyter notebook implementing the full analysis pipeline (`Crime_Analysis.ipynb`) and all generated figures accompany this report. The original dataset is publicly available from the UCI Machine Learning Repository [1].

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